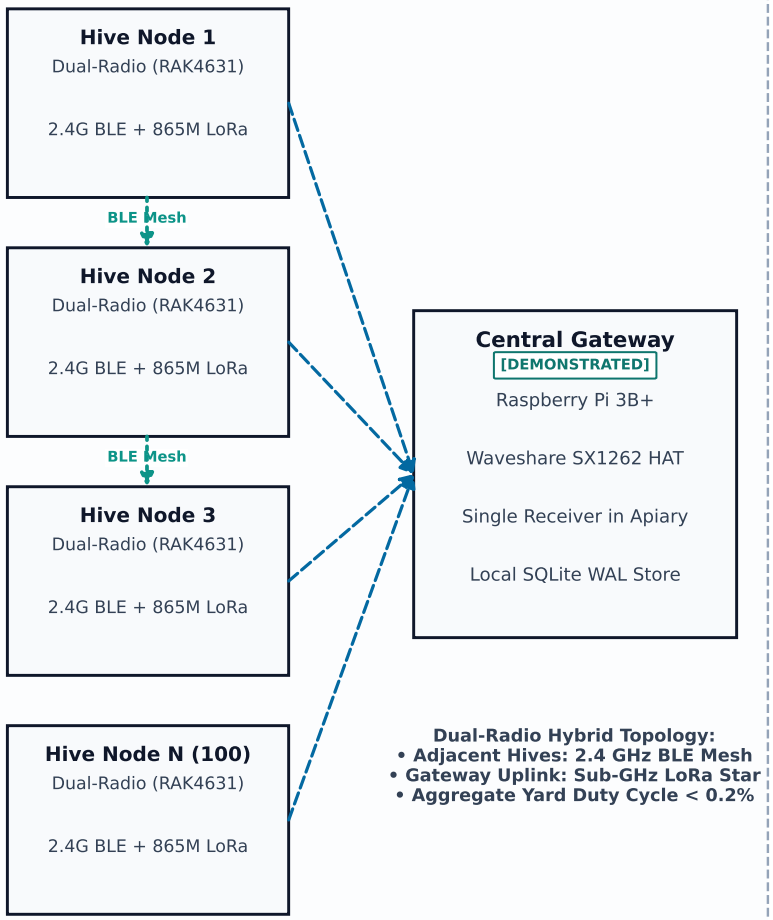
FIGURE 06: DUAL-RADIO HYBRID TOPOLOGY & LORA PACKET MEMORY MAP

2.4 GHz BLE Mesh Local Yard Cluster + Sub-GHz LoRa Star Backhaul & 33-Byte Binary Payload Structure

DUAL-RADIO HYBRID: BLE MESH CLUSTER + LORA STAR



Dual-Radio Hybrid Topology:
- **Adjacent Hives: 2.4 GHz BLE Mesh**
- **Gateway Uplink: Sub-GHz LoRa Star**
- **Aggregate Yard Duty Cycle < 0.2%**

IN865 REGULATORY LINK BUDGET & AIRTIME

RF Radio Configuration

[VALIDATED]

Carrier Frequency: 865.0625 MHz

Spreading Factor: SF7 | Bandwidth: 125 kHz

Coding Rate: 4/5 | Preamble: 8 symbols

Tx Power: +14 dBm (25 mW ERP)

Rx Sensitivity: -137 dBm

Calculated Link Performance

[CALCULATED]

Packet On-Air Time: 77.06 ms (36B v2) [CALCULATED]

Total Link Budget: 151 dB

Modelled Line-of-Sight Range [CALCULATED]

Modelled Canopy Path Loss: 1.5 km

Link Margin at 1 km: 26.16 dB

PACKED BINARY PAYLOAD MEMORY MAP (SIZEOF = 33 BYTES)

node_id	timestamp	T_core	T_grid[5]	co2_ppm	rh_c100	weight_g	bins[8]	flags	crc16
2B	4B	2B	10B	2B	2B	4B	4B	1B	2B
uint16	uint32	int16	5x int16	uint16	uint16	int32	8x 4-bit	bitmask	CCITT

Total Serialization: 33 Bytes packed struct | Zero JSON/ASCII string overhead
Payload Verification: Hardware CRC-16 CCITT polynomial verification on receiver

[MEASURED] [VALIDATED] [CALCULATED] [SIMULATED] [DEMONSTRATED]